<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>1.</td><td>(a)</td><td></td><td></td><td>Alpha – <u>helium nucleus</u> (1) Treat 2 protons and 2 neutrons as neutral
Beta – [fast moving] electron (1)
Gamma – em wave (1)</td><td>3</td><td></td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>$\frac{18}{30}$ = 0.6 [counts per second]</td><td></td><td>1</td><td></td><td>1</td><td>1</td><td>1</td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>Measure for longer (1)
Repeat (1)</td><td></td><td></td><td>2</td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>[More] radon (1)
Due to granite or different types of rock (1)

Alternative:
[More] cosmic rays (1)
Due to high altitude (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td></td><td><table><tr><td></td><td>Alpha</td><td>Beta</td><td>Gamma</td></tr><tr><td>Source 1</td><td>N</td><td>N</td><td>Y</td></tr><tr><td>Source 2</td><td>Y (1)</td><td>Y (1)</td><td>N (1)</td></tr></table>
Award 1 mark for source 1 row being correct
Accept ticks and crosses for Y and N</td><td></td><td></td><td>4</td><td>4</td><td></td><td>4</td></tr>
</table>

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Lead or concrete [absorber] should be included (1) To confirm that gamma [is emitted] (1) **OR** reduce distance between source and detector (1) To detect alpha (1) **OR** Use thick aluminium (1) To confirm no beta is emitted by source 1 (1)			2	2		2
				Question 1 total	**5**	**1**	**8**	**14**	**1**	**9**